

## FDS EXPERIMENT-10

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Experiment to understand array function in Data Science.

```
[ ]  import numpy as np

[ ] # Creating an array
arr = np.array([10, 20, 30, 40, 50])

print("Original Array:", arr)

▼ Original Array: [10 20 30 40 50]

[ ] # 1. Sum
print("Sum:", np.sum(arr))

▼ Sum: 150

[ ] # 2. Mean
print("Mean:", np.mean(arr))

▼ Mean: 30.0

[ ] # 3. Maximum
print("Maximum:", np.max(arr))

▼ Maximum: 50

[ ] # 4. Minimum
print("Minimum:", np.min(arr))

▼ Minimum: 10
```

```
# 5. Standard Deviation
print("Standard Deviation:", np.std(arr))
```

Standard Deviation: 14.142135623730951

```
# 6. Sorting
print("Sorted Array:", np.sort(arr))
```

Sorted Array: [10 20 30 40 50]

```
# 7. Reshaping
arr2 = np.array([1, 2, 3, 4, 5, 6])
print("Reshaped Array:")
print(arr2.reshape(2, 3))
```

Reshaped Array:  
[[1 2 3]  
 [4 5 6]]

```
# 8. Finding array size
print("Array Size:", arr.size)
```

Array Size: 5

```
 # 9. Finding dimensions
print("Number of Dimensions:", arr.ndim)
```

... Number of Dimensions: 1

```
# 10. Finding shape
print("Shape:", arr.shape)
```

Shape: (5,)

```
# Creating a 2D array
arr2d = np.array([[1, 2, 3],
                  [4, 5, 6],
                  [7, 8, 9]])
```

```
print("\n2D Array:")
print(arr2d)
```

2D Array:  
[[1 2 3]  
 [4 5 6]  
 [7 8 9]]



```
# Sum of all elements
print("Sum of 2D Array:", np.sum(arr2d))
```

... Sum of 2D Array: 45

```
# Row-wise sum
print("Row-wise Sum:", np.sum(arr2d, axis=1))
```

Row-wise Sum: [ 6 15 24]

```
# Column-wise sum  
print("Column-wise Sum:", np.sum(arr2d, axis=0))
```

```
Column-wise Sum: [12 15 18]
```

```
▶ # Mean  
print("Mean:", np.mean(arr2d))
```

```
... Mean: 5.0
```

```
# Maximum and Minimum  
print("Maximum:", np.max(arr2d))  
print("Minimum:", np.min(arr2d))
```

```
Maximum: 9  
Minimum: 1
```

```
# Shape  
print("Shape:", arr2d.shape)
```

```
Shape: (3, 3)
```

```
# Number of dimensions  
print("Dimensions:", arr2d.ndim)
```

```
Dimensions: 2
```

```
# Accessing an element  
print("Element at row 2, column 3:", arr2d[1, 2])
```

```
Element at row 2, column 3: 6
```